Power Electronics

Module 4

Ref: G. K. Dubey, Fundamentals of Electric Drives, Narosa publishers, second edition

Introduction to Electric Drives– Advantages of adjustable speed electric drives – Block diagram, Types of loads – Classification of load torque- Motor torque-load combination: characteristics and dynamic equation- Steady state stability [3 Hrs]

DC Drives- Chopper control of Separately Excited DC drives (SEDC) –One quadrant, Two quadrant and four quadrant Chopper fed drives (Continuous conduction only)- Motoring and Regenerative braking – Speed-Torque characteristics – Speed control- Controlled rectifier fed separately excited DC motor drive- Single phase and three phase (Continuous conduction only)- Speed-Torque characteristics- Speed control – Dual converter drives (single phase) - Circulating current Type and Non-circulating current - Static four-quadrant operation with SEDC

[5Hrs]

Three-phase VSI fed induction motor drives: Stator Voltage control - V/F speed control– Speed-Torque characteristics- Speed control – operation below and above base speed –

Braking: dynamic and regenerative [3 Hrs]

ELECTRIC DRIVE

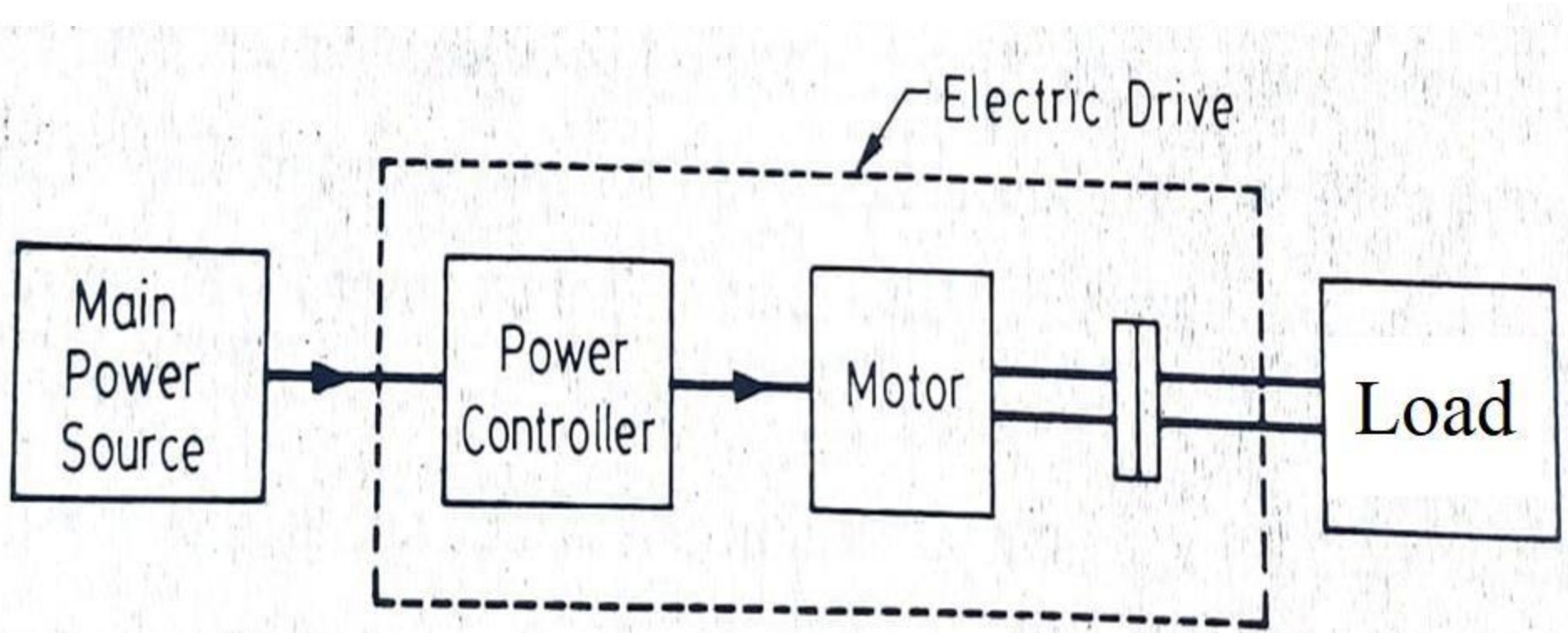


Fig. An electric-drive system.

ELECTRIC DRIVE

Electric drive:-

- is an **electromechanical device** for converting electrical energy to mechanical energy to impart motion to different machines & mechanisms for various kinds of process control.

• is characterized –

- by the nature of speed-torque characteristics such as constant speed drive or constant horse power drive.

- by the type of motor used in the drive eg.- DC & AC drives

ELECTRIC DRIVE

- **Selection of an electric drive** for a particular application is the choice of a system which provides
  - the required range of speed control most efficiently & economically.
  - desired accuracy.
  - good stability of operation.
  - good transient response.
  - reliable requiring less maintenance.

ELECTRIC DRIVE

Applications of Electric drive:-

- driving fans, ventilators, compressors, pumps etc.,
- lifting goods by hoists & cranes
- imparting motion to conveyors in factories, mines & warehouses.
- for running excavators & escalators, electric locomotives etc.

ELECTRIC DRIVE

Classification of Electric drive:-

1. depending on the ***type of supply*** – as AC drives and DC drives
2. depending on the ***number of motors used*** -3 types - as group electric drive, individual electric drive and multi-motor electric drive**.**
3. depending on the ***control schemes*** – 3 types - manual, semiautomatic and automatic.
4. depending on the ***speed*** – as variable speed drive and constant speed drive
5. depending on the ***control parameters*** – 3 types – constant power drive, constant torque drive and vector control drive

ELECTRIC DRIVES- BLOCK DIAGRAM

Drive:- is a combination of various systems combined together for the purpose of motion control or movement control.

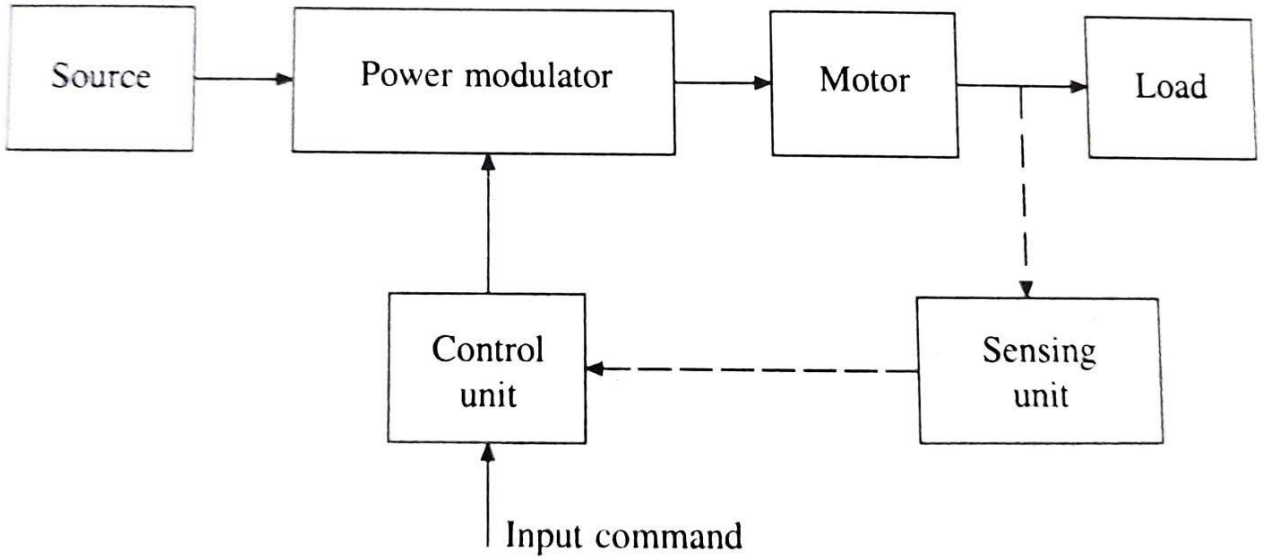
Electric Drives - the drive which employ electric motors for motion control

BLOCK DIAGRAM

- An electric motor drives the load.
- Motor is supplied from a power modulator(thyristor power converter).
- The converter has necessary controls to

provide the required supply to the motor.

For eg. Variable voltage variable frequency s



ELECTRIC DRIVES- BLOCK DIAGRAM(Contd…)

LOAD:-

- Load is usually a machinery designed to accomplish a given task. Eg.- pumps, robots, fans etc.
- Load requirements can be specified in terms of speed & torque demands.
- A motor having speed-torque characteristics & capabilities compatible or similar to load requirements are chosen.

ELECTRIC MOTORS:-

- Commonly used are

- DC Motors - shunt, series, compound and permanent magnet motors.

- AC Motors - Induction motors-squirrel cage, wound rotor(slip ring) & linear induction motor

- Synchronous motors- wound field, permanent magnet motors.

- Brushless DC (BLDC) motors, **Stepper motors** & **Switched Reluctance Motors**(SRM) are also used.

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd…)

ELECTRIC MOTORS:- (Contd…..)

- Drives using Induction & Synchronous motors
  - were used in **constant speed** drives.
  - but they are expensive
  - have very low efficiency.
- So **DC motors** started dominating in **variable speed drives** (VSDs).

Disadvantages of DC motors (with commutator & brushes) compared to AC motors:-

- High cost, weight, volume and inertia for the same rating
- Need for frequent maintenance
- Unsuitable for explosive and contaminated environment
- Restrictions on maximum voltage, speed and power ratings.

Due to development of **semiconductor devices** like thyristors, IGBTs, power transistors etc., AC motors are widely used in VSDs.

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

ELECTRIC MOTORS:- (Contd…..)

AC MOTORS:-

Squirrel cage induction motors:-

- Cost is one-third of a DC motor of same rating.
- Rugged, requires no maintenance.
- Can be built for higher speeds, torques and power ratings.

Wire wound motors:-

- More expensive than squirrel cage induction motors.
- Less maintenance than DC motors but more than squirrel cage induction motors.
- Available in high power ratings.

Wound field and permanent magnet synchronous motors(PMSM):-

- Higher full load efficiency and power factor than induction motors.
- Can be designed for higher power rating than induction motors.
- Higher cost and size for same rating.
- Require more maintenance compared to squirrel cage induction motors.
- PMSMs have all the advantages of squirrel cage induction motors but the power rating is low.

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

ELECTRIC MOTORS:- (Contd…..)

Brushless DC motors (BLDC Motors):-

- Similar to PMSM , but has low cost.
- Requires simpler and cheaper converter.
- Considered for low power high speed drives
- Used for servo applications as an alternative to DC servo motor.

DC servo motor:-

- Has all the disadvantages of DC motors with commutators and brushes.
- At low power levels, the friction between brushes and commutators is objectionable. It affects the steady state accuracy of the drive.

Stepper motors:-

- Popular for position control

Switched reluctance motor(SRM):-

- Popular for speed control

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

POWER MODULATORS:-

Functions:-

(i) **Modulates the flow of power** from source to the motor in such a manner that the motor gives a speed-torque characteristics as required by the load.

(ii) During transient operation, like starting, braking and speed reversal, the power modulator **limits the source and motor currents** within the permissible values. Excessive currents drawn from the source may overload it and may cause a voltage dip.

(iii) Converts electrical energy of the source to a form suitable to the motor.

(iv) Selects the mode of operation of motor. i.e., motoring or braking.

When the power converter is used to perform the third function it is called a **CONVERTER**. Depending on its circuit, a converter may also perform other functions also. A converter is needed when the nature of the available electrical power is different than what is required for the motor.

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd…)

POWER MODULATORS:-

Classification:-

(i) Converter

(ii) Variable impedance

(iii) Switching circuits

(i) Converters:-

1. Rectifiers (AC to DC converters)
2. AC regulators (AC to AC converters)
3. Choppers (DC to DC converters)
4. Inverters (DC to AC converters)
5. Cycloconverters (AC to AC converters)

Power sources are of two types:-

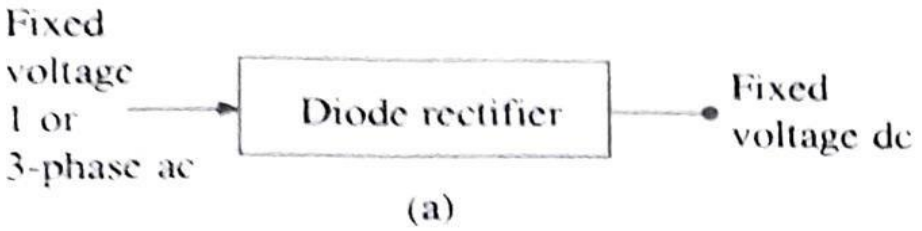
- fixed voltage fixed frequency AC
- fixed voltage DC

For DC motor control variable DC voltage and for AC motor control variable voltage AC or variable voltage variable frequency AC is required. These requirements are met by the converters.

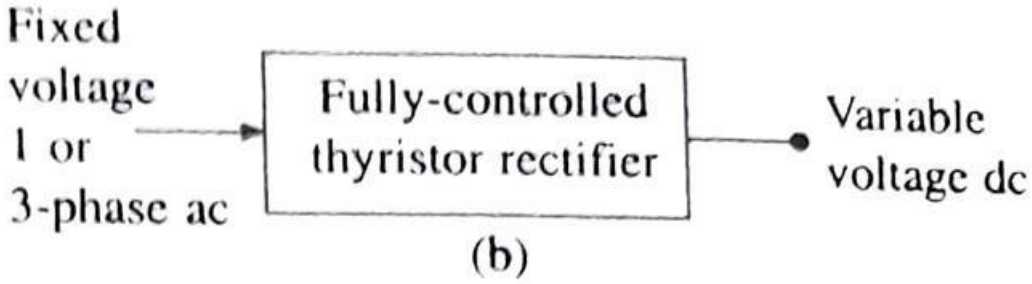
ELECTRIC DRIVES- BLOCK DIAGRAM(Contd…)

1. <u>AC to DC converters:-</u>

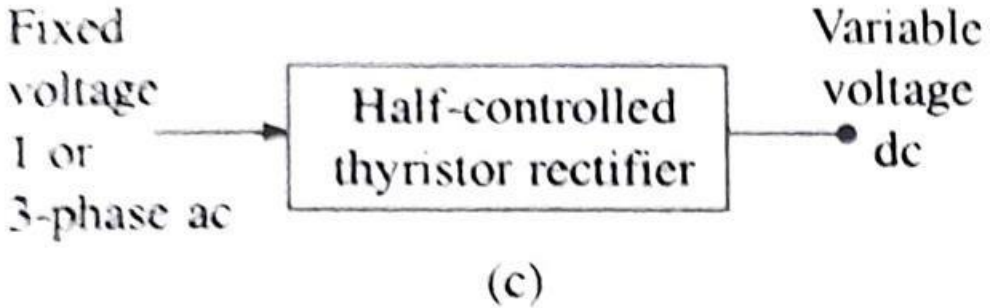
- (a) **Uncontrolled rectifier**



- (b) **Controlled rectifier-** two quadrant converter -capable of providing variable DC voltage of either polarity with +ve current - produce harmonics both on DC & AC side- have low power factor for DC voltages



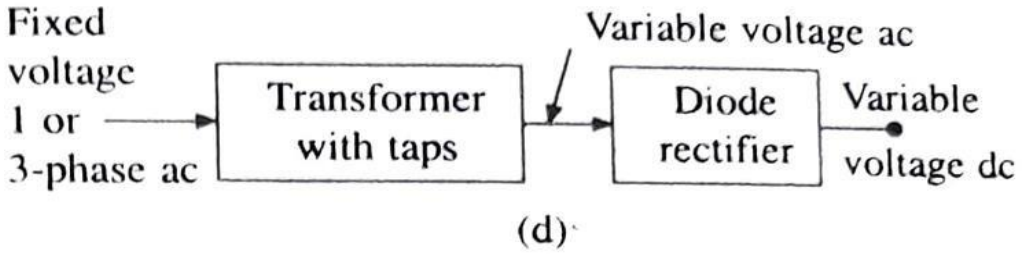
- (c) **Controlled rectifier** – is a single quadrant converter - +ve voltage & +ve current- produce harmonics both on DC & AC side- have low power factor for DC voltages



ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

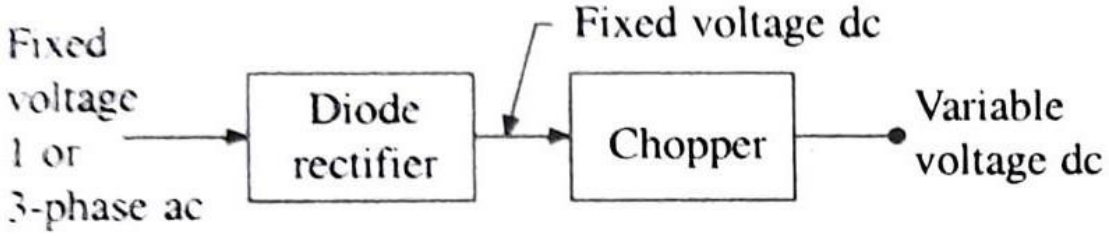
1. AC to DC converters :- (Contd…..)

• (d) **Uncontrolled rectifier** – operate at unity fundamental power factor- o/p voltage is changed by applying a mechanical force- few discrete steps of DC voltage can only be obtained



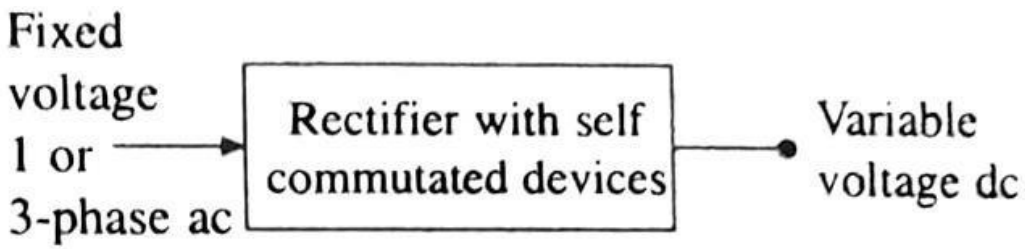
(d)

• (e) - operate at unity fundamental power factor
- Output voltage can be varied in a step less manner by controlling the duty ratio of the semiconductor device of the chopper by low power electrical signals from a control unit.



(e)

• (f) **Controlled rectifier** – employing semiconductor devices power transistors, IGBTs & GTOs-can be single quadrant or 2 quadrant depending on the circuit -operate at unity fundamental power factor

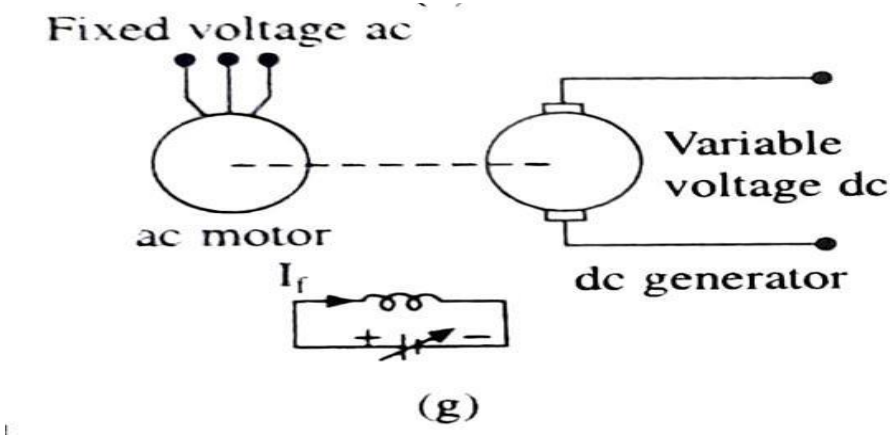


(f)

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

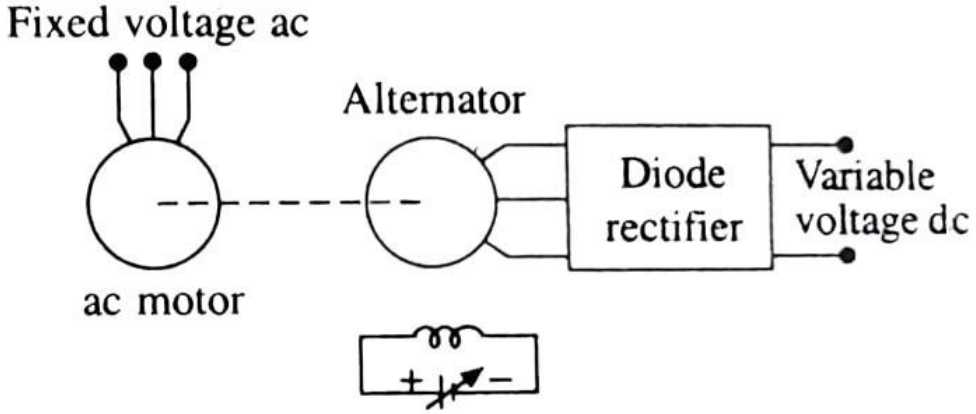
1. AC to DC converters :- (Contd…..)

- (g)**Uncontrolled rectifier**- the output voltage is controlled by controlling the field current of the generator from a control unit of higher power level- can operate in all 4 quadrants- number of disadvantages because of 2 rotating machines- bulky, heavy, noisy, less efficient, slow response, expensive, requires special foundation.



(g)

- The converter in fig (h) can operate in single quadrant only

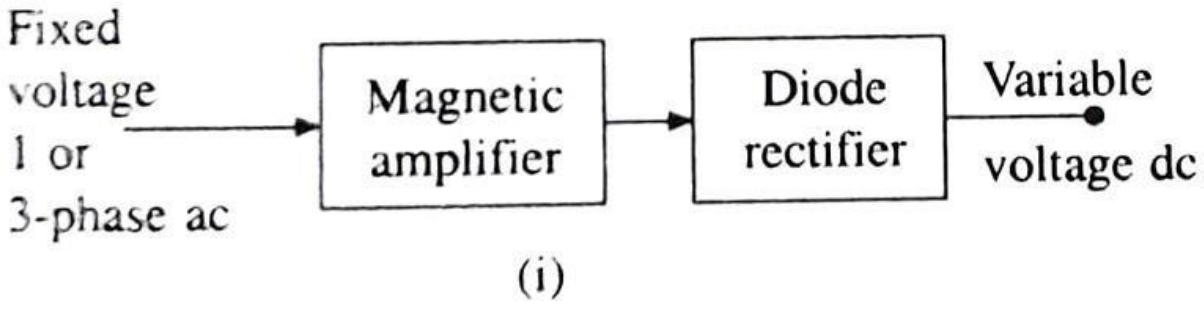


(h)

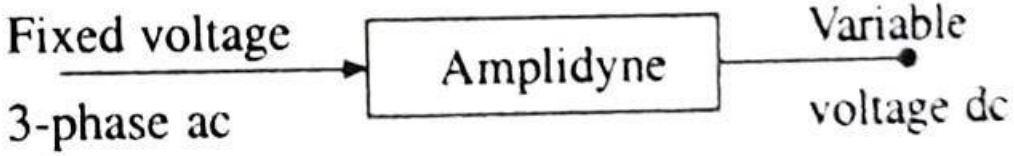
ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

1. AC to DC converters :- (Contd…..)

- Some very old equipments may use converters in fig (i) & fig (j) - magnetic amplifiers & amplidynes are controlled from low power DC signals



(i)

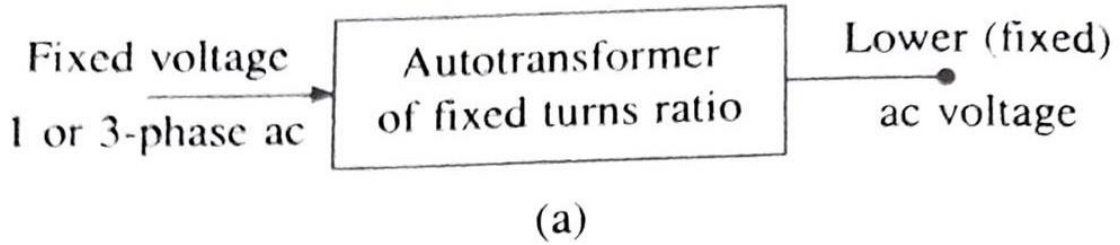


(j)

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

2. AC voltage controllers :-

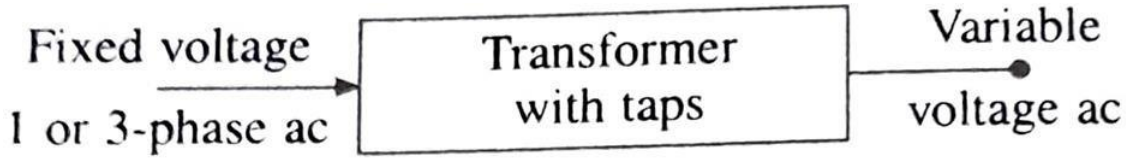
- (a)



(a)

- Gives fixed AC voltage.
- Autotransformers capable of giving variable output voltage are not employed due to sliding contacts

- (b)



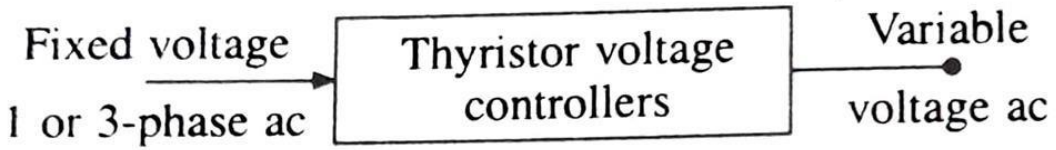
(b)

- gives variable AC voltage with few discrete steps.
- control is exercised by a mechanical switch.
- output voltage and current are sinusoidal.

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

2. AC voltage controllers :- (Contd…..)

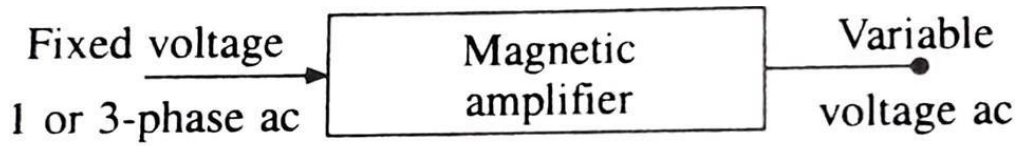
- (c)



(c)

- a thyristorised voltage controller is used here
- stepless control of output voltage can be obtained by controlling the firing angle of converter.
- output voltage and source current have harmonics
- poor power factor at low output voltages.

- (d)

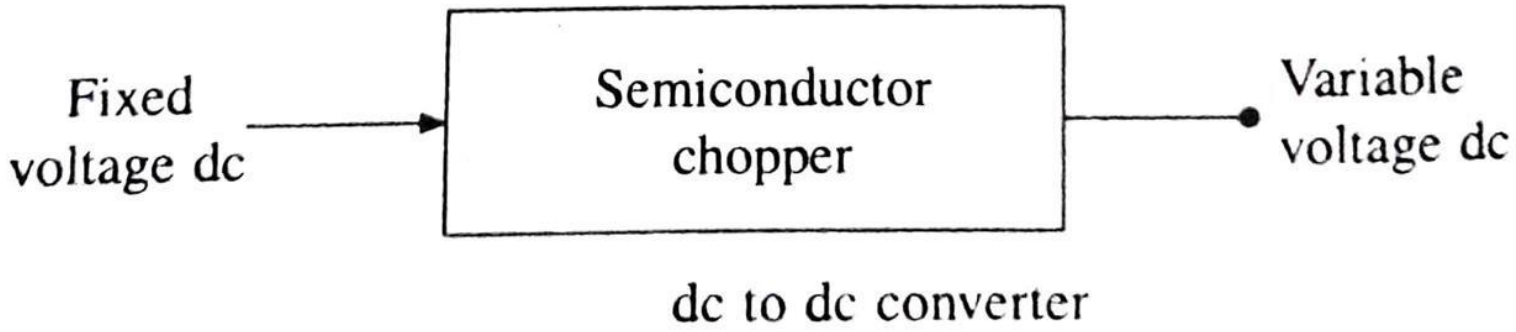


(d)

- old drives employ magnetic amplifier
- high cost, weight and volume and poor efficiency – so replaced by thyristor controllers.

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

3. Choppers:

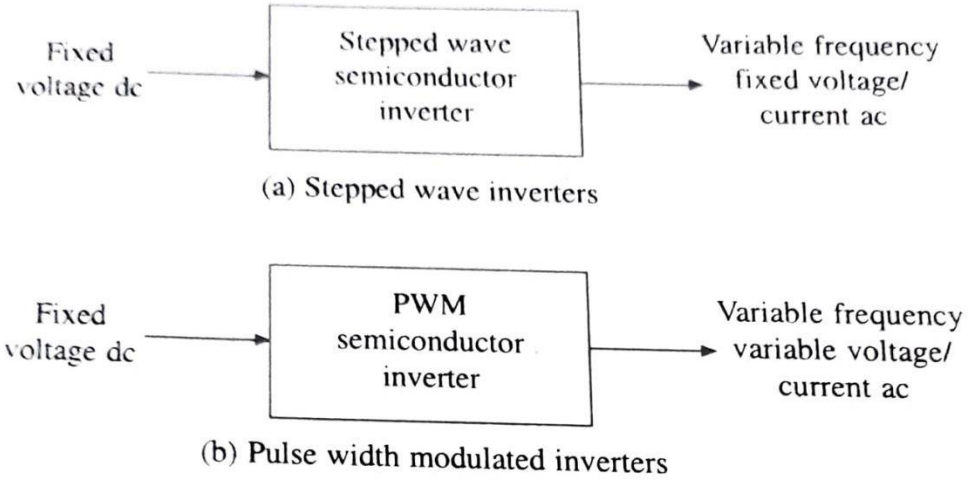


- Used to get variable DC voltage from a fixed DC voltage
- Uses semiconductor devices like power transformers, IGBTs, power MOSFET, thyristors etc.
- Output voltage can be varied in a step less manner by controlling the duty ratio of the chopper.

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

4. Inverters:-

- To get variable frequency AC from a fixed DC supply.
- Stepped wave inverters- can be designed to behave as voltage source or current source and they are known as voltage source or current source inverters.

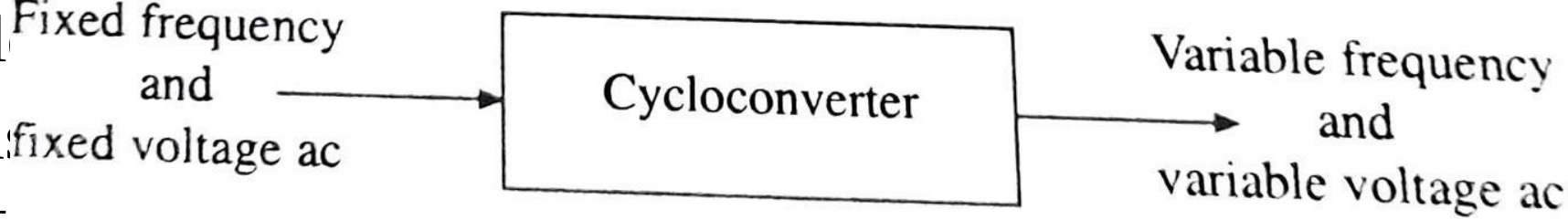


- For AC motor control, voltage/current should also be controlled along with frequency.
- Variation in output voltage/current can be achieved by varying the input DC voltage. This is achieved by placing a chopper in between fixed DC source and the inverter.
- Output contains considerable harmonics.
- When the inverter is controlled by PWM, variable frequency variable voltage AC is directly obtained from fixed voltage DC.
- PWM reduces harmonics in output voltage
- Built of semiconductor devices like thyristors, IGBTs, GTOs etc., are controlled by controlling firing angle.

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

5. Cycloconverters:-

- Converts fixed voltage fixed frequency AC to variabl
- Built u
- Output frequency is restricted to 40% of supply frequency to keep harmonics in output voltage and source currents within limits.



ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

POWER MODULATORS - Classification :- (Contd…..)

(ii) <u>Variable impedance:-</u>

- Variable resistors

  - commonly used for the control of low cost DC & AC drives.

  - needed for the dynamic braking of drives.

  - can be controlled manually or automatically with the help of contactors.

- stepless variation of resistance can be obtained using a semiconductor switch in parallel with a fixed resistance.

  - variation of duty ratio of the switch gives a stepless variation in effective value of the resistance.

- Variable inductors

  - usually in two steps(full or zero), are employed

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

POWER MODULATORS - Classification :- (Contd…..)

(iii) Switching circuits:-

Switching operations are required to achieve:-

- For changing motor connections to change its quadrant of operations
- For operating motor circuit parameters in discrete steps for automatic starting and braking control.
- For operating motors and drives according to a predetermined sequence.
- To prevent maloperation.
- To disconnect motor when abnormal operating conditions occur.

Switching operations in motor's power circuit are done by high power electromagnetic relays known as **contactors.**

Thyristor switches – disadvantages- no isolation between source and motor circuit. So contactors are still used.

Low power electromagnetic relays were used for sequencing & interlocking operations. Nowadays solid state relays are used.

Programmable logic controllers(PLCs) are employed to implement complicated sequencing & interlocking operations.

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

SOURCES:-

- **Low power drives** are fed from single-phase source.
- **High power drives** are fed from three-phase source except traction drives where single-phase supply is used even at very high power levels because of economy.
- When fed from 50Hz AC, the maximum speeds of induction motor and synchronous motors are limited to 3000 rpm.
- **Low and medium power motors** – fed from 400V supply

  High power motors – fed from 3.3 kV, 6.6 kV, 11 kV and higher are used
- **Aircraft & space applications** – 400 Hz AC supply is used- to achieve high power to weight ratio.
- **Main line traction** – high voltage supply is used- because of economy- in India 25 kV 50 Hz supply is used.
- **Drives powered from battery** – fork lift trucks(is a powered industrial truck used to lift and move materials over short distances), milk vans etc. Battery voltage may be typical values of 6 V, 12 V, 24 V, 48 V and 110 V DC.
- Drives powered from low voltage DC supply- **solar powered drives-** used in space & water pumping applications.

ELECTRIC DRIVES- BLOCK DIAGRAM(Contd….)

CONTROLUNIT:-

- Controls for power modulator is provided in this unit.
- Nature of the control unit depends on the power modulator used.
- When semiconductor converters are used, the control unit consists of firing circuits.
- When control of switching circuits is required, the control unit will provide sequencing and interlocking.
- Solid state relays are also used but when control is complex PLCs are used.

SENSING UNIT:-

Performs **two functions**:-

- **Speed sensing**

  - required for implementation of closed loop control schemes.

  - speed sensed by tachometers.

- when high speed accuracies are required digital tachometers are used. Eg.- paper mills, computer peripherals.

- **Current sensing**- employs two methods

  - use of current sensor using hall effect.

- use of non-inductive resistance shunt in conjunction with an isolation amplifier- has an arrangement for an amplification and isolation between power and control circuits.

ELECTRIC DRIVES

ADVANTAGES:-

- Have **flexible control characteristics**. The steady state and dynamic characteristics can be shaped to satisfy load requirements.
- Automatic fault detection systems, PLCs and computers can be used to **automatically control the drive operations** in desired sequence.
- Available in **wide range of torque, speed and power**.
- **Can operate in all the 4 quadrants** of speed-torque characteristics. Electric braking gives smooth deceleration and increases life of the equipment compared to other forms of braking.
- Can be used in **any operating conditions** like explosive and radioactive environments and electrical drives do not pollute the environment.
- There is **no need to re-fuel** the motor.
- Can be **started instantly** and can be **fully loaded immediately**.
- Powered by **electrical energy which has a number of advantages** over other forms of energy. It can be generated and transported to the desired point economically and efficiently.

ELECTRIC DRIVES

Comparison of DC and AC drives:-

DC drives	AC drives
1. Commutator makes the motor bulky, costly and heavy.	Motors are inexpensive, particularly squirrel cage motor.
2. Line commutation of the converter is used.	Forced commutation is used with induction motors.
3. Poor power factor, harmonic distortion of the current.	For regenerative drives line power factor is poor.
4. Fast response and wide speed range and smooth control.	Response depends on the type of control. With solid state converters speed range is wide. With conventional methods it is stepped and limited
5. Small power/weight ratio.	Large power/weight ratio.
6. Sparking at the brushes makes it unsuitable for certain locations.	Problems like sparking at the brushes are not here.
7. Highest speed and design rating are limited due to commutation. Commutator requires frequent maintenance.	No upper limit for speed and design rating. Motor is reliable , requires less maintenance, can be used in all locations.

ELECTRIC DRIVES

Choice of electrical drive depends on :-

1. **Steady state operation requirements**

- nature of speed-torque characteristics, speed regulation, speed range, efficiency, duty cycle, quadrants of operation, speed fluctuations.

2. **Transient operation requirements**

- values of acceleration and deceleration, starting, braking and reversing performance.

3. **Requirements related to the source**

- type of source, capacity, magnitude of voltage, voltage fluctuations, power factor, harmonics, ability to accept regenerated power.

4. Capital and running cost, maintenance needs, life

5. Space and weight restrictions if any.

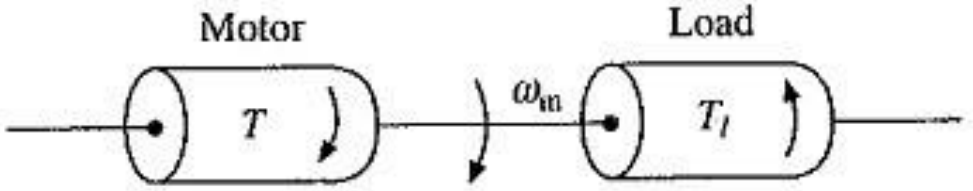
6. Environment and location

7. Reliability

DYNAMICS OF ELECTRIC DRIVES

FUNDAMENTAL TORQUE EQUATION

- A motor drives a load (or a machine) through some transmission system.
- The motor always rotates but the load may rotate or may undergo translational motion.
- The speed of the load may be different from the speed of the motor.
- If the load has many parts-their speeds may be different-some may rotate-some may go through translational motion.
- Fig shows the representation of motor-load system by an equivalent rotational system.



Equivalent motor-load system

FUNDAMENTAL TORQUE EQUATION(Contd….)

- Let

J = polar moment of inertia of motor-load system referred to motor shaft; kg-m^2.

ω_m= instantaneous angular velocity of the motor shaft; rad/sec.

T =instantaneous value of developed motor torque; N-m.

T_l =instantaneous value of load torque referred to motor shaft; N-m.

- The developed motor torque is considered as an applied torque
- Load torque includes friction & windage torque of the motor. It is also called resisting torque.
- The **fundamental torque equation** is

$$T - Tl = \frac{d}{dt}(J\omega_m) = J\frac{d\omega_m}{dt} + \omega_m \frac{dJ}{dt} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

FUNDAMENTAL TORQUE EQUATION(Contd….)

- Eqn (1) is applicable to variable inertia drives like mine winders, reel drives, industrial robots etc.
- For drives with constant inertia $\frac{dJ}{dt} = 0$
- The eqn (1) becomes

$$T - T_l = J \frac{d\omega_m}{dt}$$

$$or \qquad T = T_l + J \frac{d\omega_m}{dt} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

= load torque + dynamic torque

- Torque developed by motor is counterbalanced by load torque & a dynamic torque.
- $J\frac{d\omega_m}{dt}$ - represents inertia torque-is called **dynamic torque-** is present only during transient operations, i.e., when the speed of the drive varies.
- During acceleration of the drive, inertia torque is directed against the motion.
- During braking the inertia torque maintains the motion of the drive.

CLASSIFICATION OF LOAD TORQUE

- **Classification of loads**:-

  - those which can provide active torques called active loads

  &

  - those which can provide passive torques called passive loads

- **Active load torque**:-

  - due to either gravitational force or deformation in elastic bodies.

- i.e., torque due to the force of gravity and torques due to tension, compression and torsion undergone by an elastic body come under this category.

  - are the load torques which have the potential to drive the motor under equilibrium conditions.

  - these torques retain their sign when the direction of drive is changed.

  - obtained in hoists, lifts or elevators & railway locomotives operating on gradients

  - developed during compression or release of springs

- the function of hoists, operation of locomotives on gradients, compression or release of springs are all associated with a change in potential energy of the drive. Thus active torques are also connected to potential energy.

CLASSIFICATION OF LOAD TORQUE(Contd…)

- **Passive torques:-**

- load torques which always oppose the motion and change their sign on the reversal of motion

  - due to friction or due to shear & deformation in inelastic bodies

- always oppose motion- thus retards the rotation of the driven machine.

- when there is a change in the direction of motion, the direction of torque also changes.

Eg:- when a weight is lifted, the friction torque gets added up to the useful torque and when the weight is lowered the friction torque gets subtracted from the useful torque.

CLASSIFICATION OF LOAD TORQUE(Contd…)

- when a load is moved upward or a spring is compressed (there is a vertical movement, therefore increase in PE), the stored potential energy increases. The active torque developed opposes the action that takes place. i.e., the torque is directed against the upward movement or compression.

- when a load is brought downward or a spring is released, the stored potential energy decreases and the developed torque aids this action. Thus the torque continue to act in the same direction even after the direction of the drive is reversed.

COMPONENTS OF LOAD TORQUE

- Load torque(T_l) can be further divided into:-
  - friction torque – T_F
  - windage torque – T_w
  - torque required to do useful mechanical work- T_L

COMPONENTS OF LOAD TORQUE(Contd…)

- **Friction torque – T_F :-**

- friction will be present at the motor shaft &various parts of the load.

- is the equivalent value of various friction torques referred to the motor shaft.

- 3 components – T_v , T_c and T_s.

-viscous friction T_v - varies linearly with speed is called viscous friction and $T_v = B\omega_m$,……(3)

where B= viscous friction coefficient.

- **Coulomb friction –** Tc – independent of speed

- T_s – is the additional torque present at standstill – present only at standstill- so not taken into account for dynamic analysis.

- Friction at zero speed is called **static friction** or **stiction**.

- For a drive to start, the motor torque should at least exceed static friction.

COMPONENTS OF LOAD TORQUE(Contd…)

- **Windage torque – T_w :-**

  - it is the opposition offered by air to the motion.

- when a motor runs, wind generates a torque opposing the motion

  - proportional to square of the speed.

  - $T_w = C\omega_m^2$. ……..(4)

    where C is a constant.

COMPONENTS OF LOAD TORQUE(Contd…)

- Torque required to do useful mechanical work- $\boldsymbol{T_L}$

  - nature of this torque depends on the type of the load.

- may be constant and independent of speed or may be some function of speed.

  - may depend on the position or path followed by load.

  - may be time invariant or time variant.

  - may vary cyclically.

  - may change with the load's mode of operation.